



南方科技大学  
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

考试科目: 线性代数精讲

开课单位: 数学系

考试时长: 110 分钟

命题教师: 线性代数精讲教师团队

| 题号 | 1    | 2    | 3    | 4    | 5    | 6    | 7    |
|----|------|------|------|------|------|------|------|
| 分值 | 30 分 | 10 分 | 10 分 | 20 分 | 20 分 | 10 分 | 10 分 |

本试卷共 ( 7 ) 大题, 满分 ( 110 ) 分. (考试结束后请将试卷、答题本、草稿纸一起交给监考老师)

This test includes 7 questions. Write **all your answers** on the examination book.

Please put away all books, calculators, cell phones and other devices. Please write carefully and clearly in complete sentences. Your explanations are your only representative when your work is being graded.

Unless otherwise noted, vector spaces are over  $\mathbb{F}$  and with finite dimensions, where  $\mathbb{F} = \mathbb{R}$  or  $\mathbb{F} = \mathbb{C}$ .

1. (30 points, 6 points each) Label the following statements as **True** or **False**. **Along with your answer, provide an informal proof, counterexample, or other explanation.**

- (a) Suppose  $U_1, U_2, U_3$  are subspaces of  $V$  such that  $U_1 + U_2 = U_2 + U_3$ , then  $U_1 = U_3$ .
- (b) Suppose  $v_1, v_2, \dots, v_n$  is linearly independent, then  $v_1, v_1 + v_2, \dots, v_1 + v_2 + \dots + v_n$  is linearly independent.
- (c) Suppose  $P \in \mathcal{L}(V)$  and  $P^2 = P$ . Then  $V = \text{null } P \oplus \text{range } P$ .
- (d)  $x^3 - 2x^2 + 1, 4x^2 - x + 3, 3x - 2, 1$  spans  $\mathcal{P}_3(\mathbb{R})$ .
- (e) Let  $V$  be a finite-dimensional real inner product space. If  $v_1, v_2, \dots, v_m \in V$ , then

$$\sum_{i=1}^m \sum_{j=1}^m \langle v_i, v_j \rangle \geq 0.$$

2. (10 points) Suppose  $\lambda_1, \lambda_2, \dots, \lambda_n$  is a list of distinct real numbers. Prove that the list

$$e^{\lambda_1 x}, e^{\lambda_2 x}, \dots, e^{\lambda_n x}$$

is linearly independent in the vector space of real-valued functions on  $\mathbb{R}$ .

3. (10 points) Suppose  $V$  is a finite-dimensional vector space and  $U_1, U_2, U_3, U_4$  are subspaces of  $V$ . Show that:

$$U_1 + U_2 + U_3 + U_4$$

is a direct sum **if and only if**

$$\dim(U_1 + U_2 + U_3 + U_4) = \dim U_1 + \dim U_2 + \dim U_3 + \dim U_4.$$

4. (20 points) Let  $A = \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix}$  and  $T$  be the linear transformation from  $\mathbb{R}^{2 \times 2}$  to  $\mathbb{R}^{2 \times 2}$  defined by

$$T(X) = XA + AX, \quad X \in \mathbb{R}^{2 \times 2}.$$

Where  $\mathbb{R}^{2 \times 2}$  denotes the vector space consisting of all  $2 \times 2$  real matrices.

- (a) Find the matrix representation of  $T$  with respect to the following ordered basis

$$v_1 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}, \quad v_3 = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}, \quad v_4 = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}.$$

- (b) Show that  $T$  is invertible.

- (c) Find an  $S \in \mathcal{L}(\mathbb{R}^{2 \times 2})$ , such that

$$ST = TS = I.$$

5. (20 points) Let  $V$  and  $W$  be two finite dimensional vector spaces and  $T \in \mathcal{L}(V, W)$ . Suppose  $v_1, v_2, \dots, v_n$  is a basis of  $V$  and  $w_1, w_2, \dots, w_m$  is a basis of  $W$ . The matrix of  $T$  with respect to  $v_1, v_2, \dots, v_n$  and  $w_1, w_2, \dots, w_m$  is  $A$ . Show that:

$$\text{rank } A = \dim \text{range } T.$$

6. (10 points) Suppose  $V$  and  $W$  are finite-dimensional and  $T \in \mathcal{L}(V, W)$ .

- (a) Suppose  $U$  is a subspace of  $V$ . Show that  $U^\circ = \{0\}$  if and only if  $U = V$ .

- (b) Show that  $T$  is surjective if and only if  $T'$  is injective.

7. (10 points) This problem has two parts.

- (a) Let  $\alpha_1, \alpha_2, \dots, \alpha_p$  be linearly independent  $m$ -dimensional column vectors and  $\beta_1, \beta_2, \dots, \beta_q$  be linearly independent  $n$ -dimensional column vectors. Show that

$$\alpha_i \beta_j^T \quad (i = 1, 2, \dots, p, \quad j = 1, 2, \dots, q)$$

is linearly independent.

- (b) Consider the following linear operator:

$$T(X) = AXA, \quad X \in \mathbb{R}^{n \times n}.$$

Suppose  $A$  is diagonalizable, show that  $T$  is diagonalizable.